

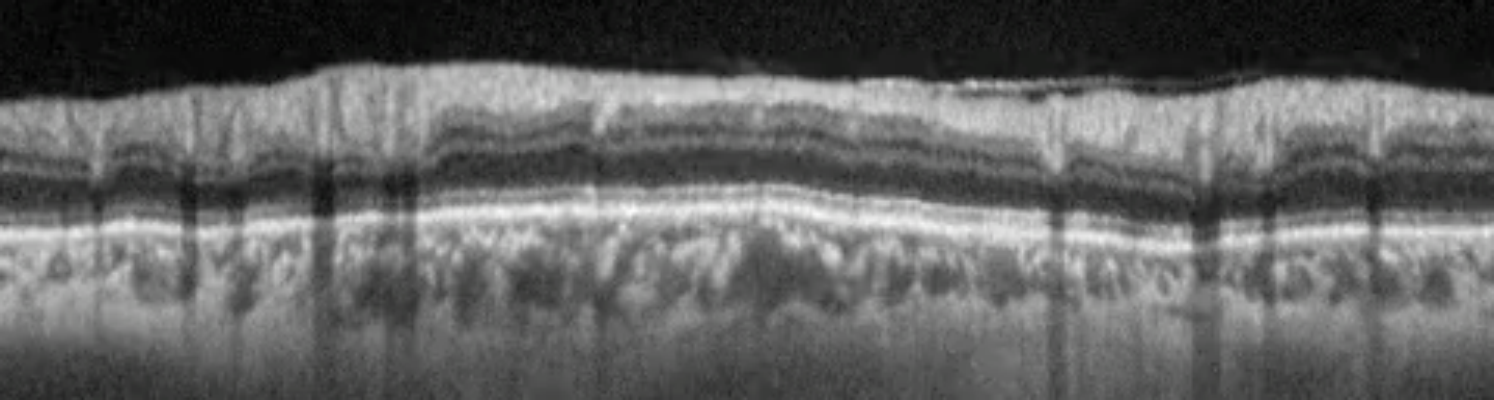
Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

RWTHAACHEN
UNIVERSITY

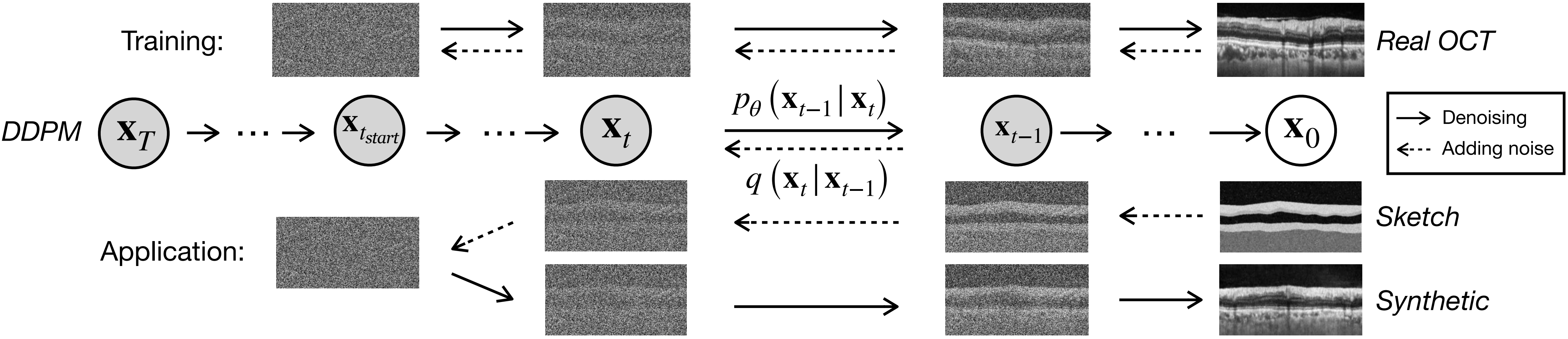
4

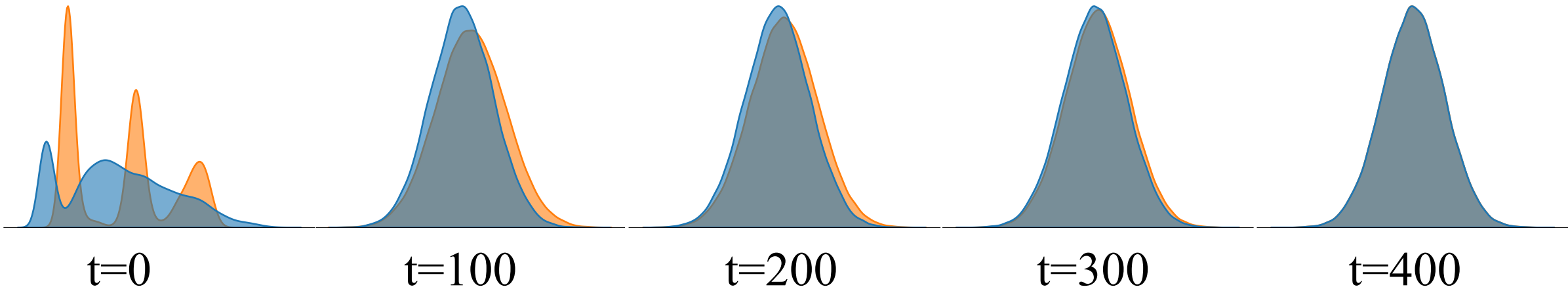
2

OCISynthesis for Layer Segmentation







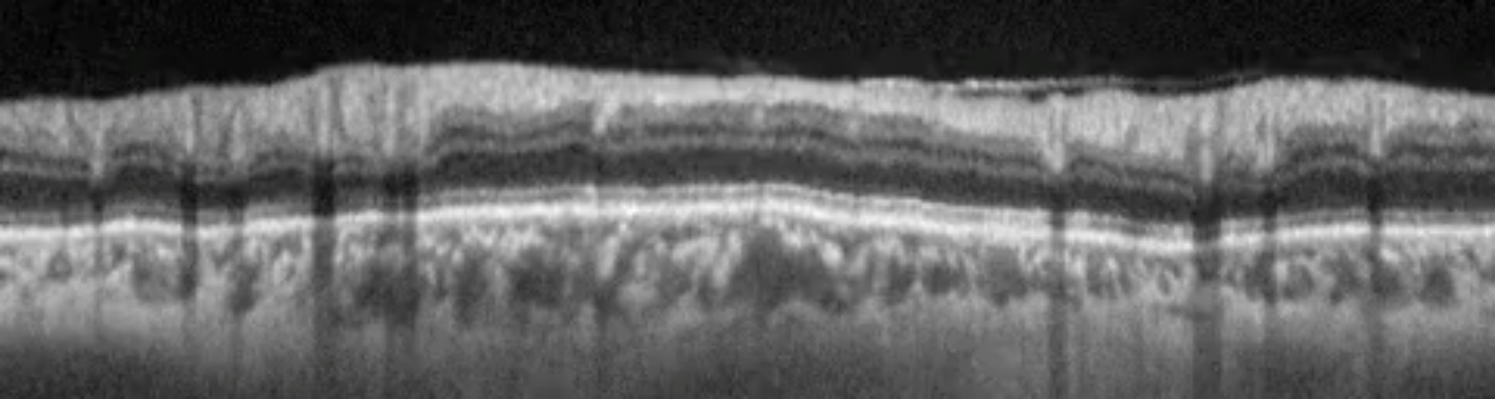


$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

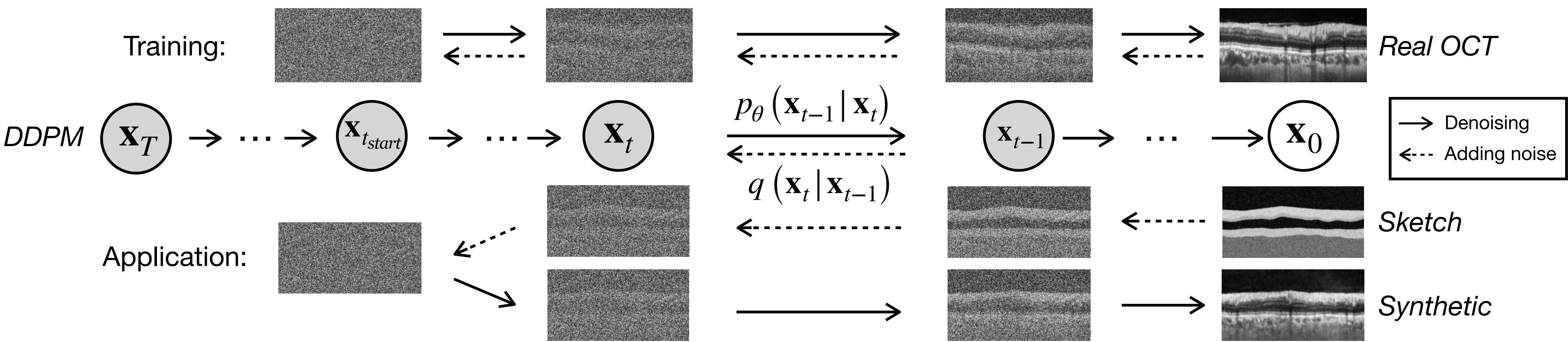
$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

[Wuetai., 2024]



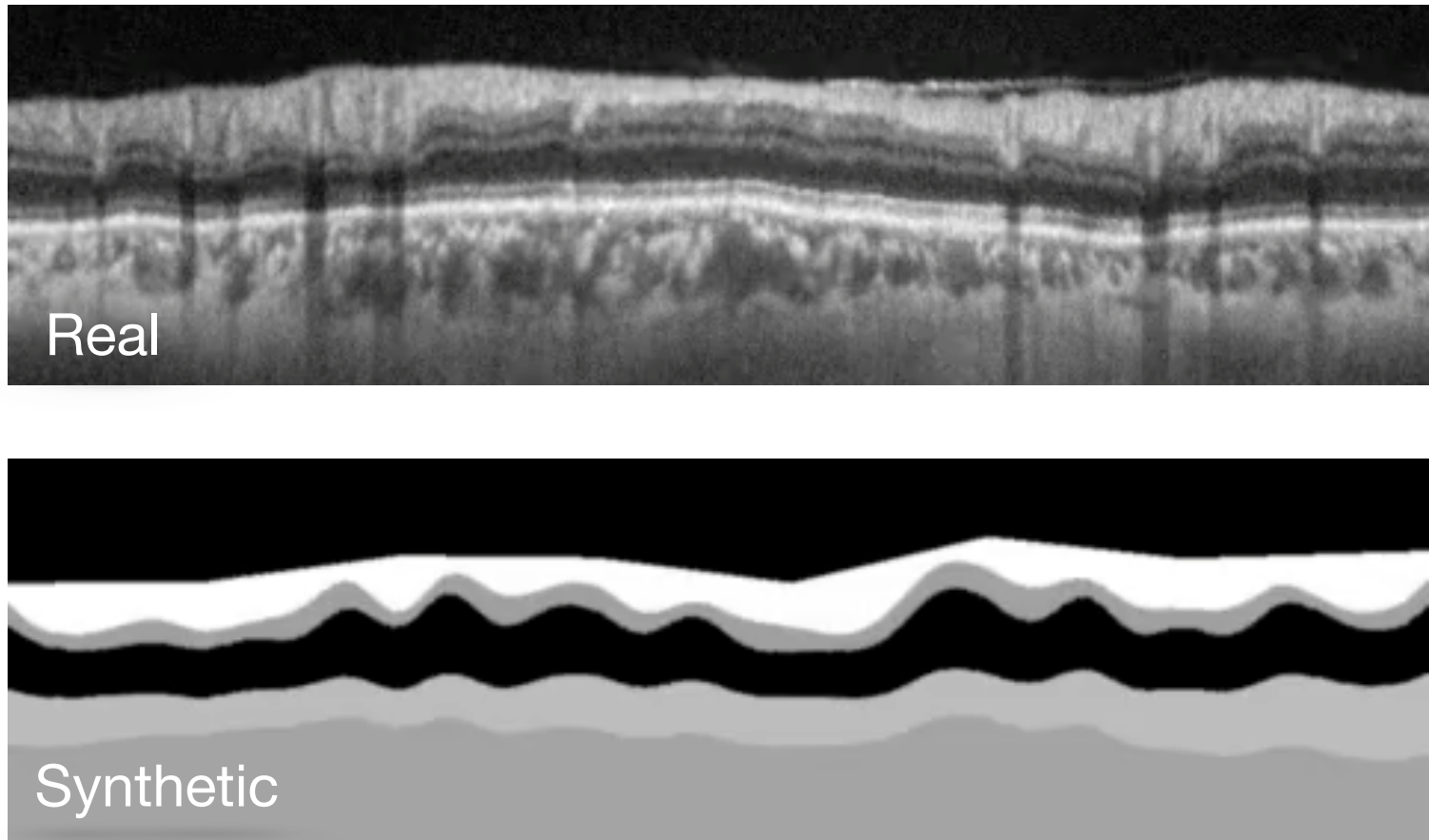
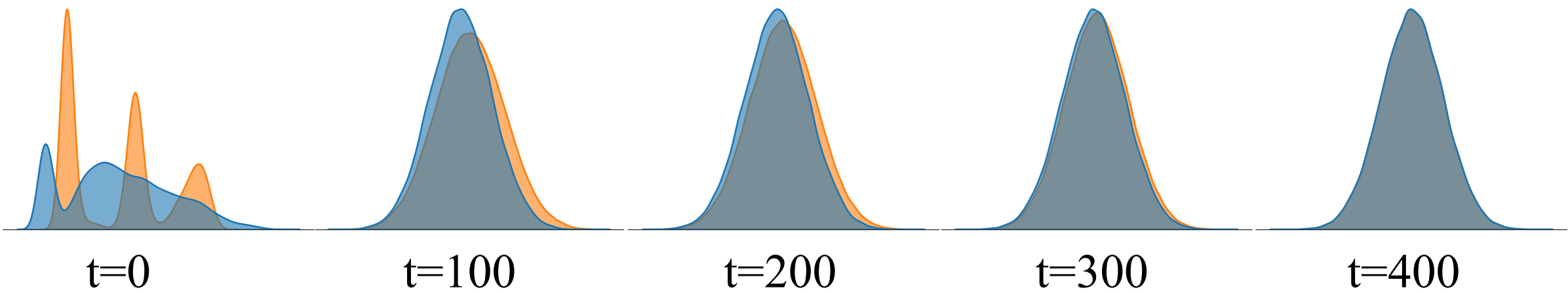


OCT Synthesis for Layer Segmentation



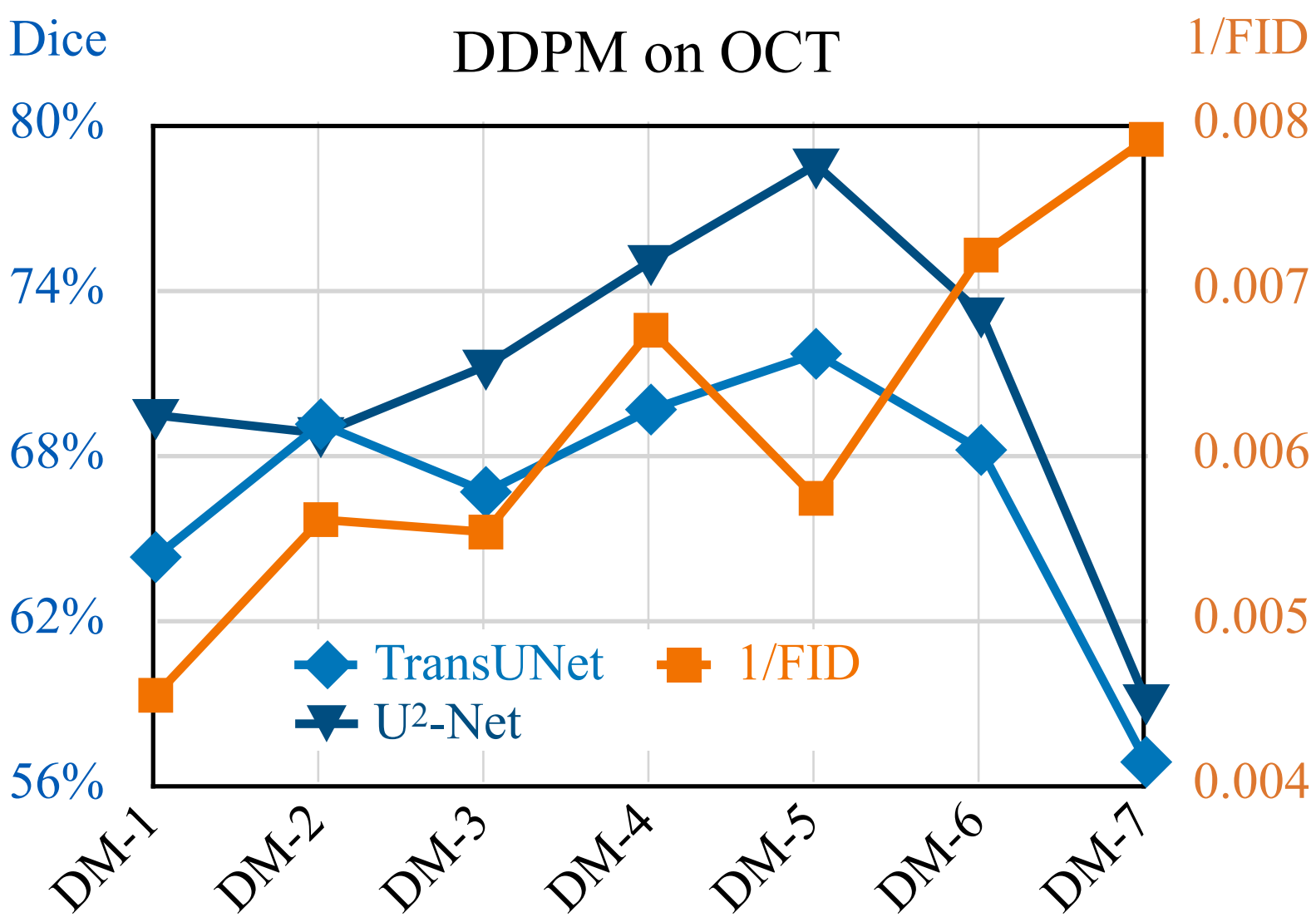
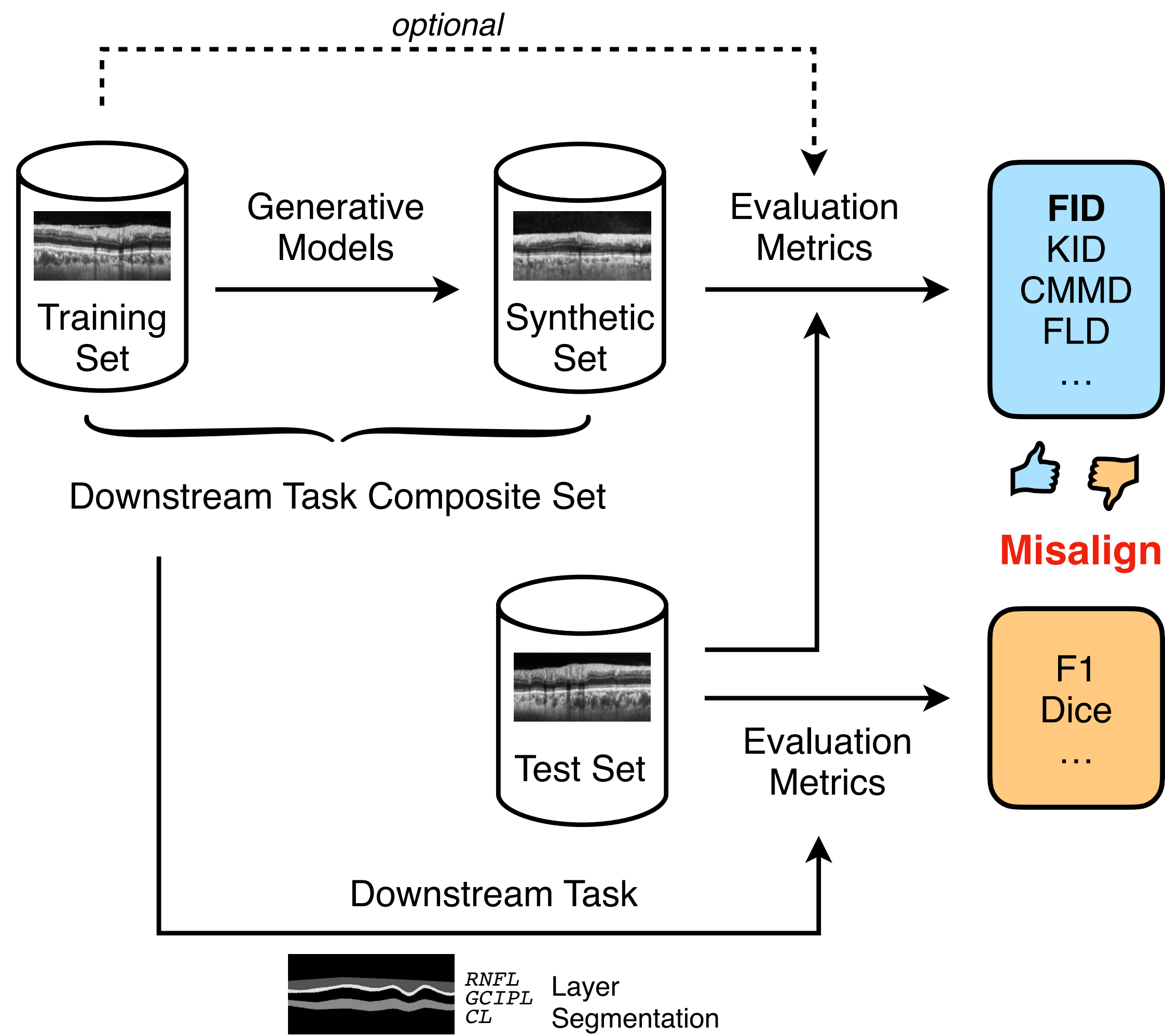
$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t))$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$



[Wu et al., 2024a]

Evaluating Generative Models



DDPM on OCT

FID	1.00	1.00 ***	0.81 *	0.52 n.s.	1.00 ***	0.71 *	0.81 *	0.05 n.s.	-0.05 n.s.	-0.14 n.s.
Clean-FID	1.00 ***	1.00	0.81 *	0.52 n.s.	1.00 ***	0.71 *	0.81 *	0.05 n.s.	-0.05 n.s.	-0.14 n.s.
CLIP-FD	0.81 *	0.81 *	1.00	0.71 *	0.81 *	0.71 *	0.62 n.s.	-0.14 n.s.	-0.05 n.s.	-0.14 n.s.
RETF.-FD	0.52 n.s.	0.52 n.s.	0.71 *	1.00	0.52 n.s.	0.62 n.s.	0.33 n.s.	-0.24 n.s.	-0.14 n.s.	-0.24 n.s.
KID	1.00 ***	1.00 ***	0.81 *	0.52 n.s.	1.00	0.71 *	0.81 *	0.05 n.s.	-0.05 n.s.	-0.14 n.s.
CMMD	0.71 *	0.71 *	0.71 *	0.62 n.s.	0.71 *	1.00	0.52 n.s.	0.14 n.s.	0.05 n.s.	-0.05 n.s.
FLD	0.81 *	0.81 *	0.62 n.s.	0.33 n.s.	0.81 *	0.52 n.s.	1.00	-0.14 n.s.	-0.24 n.s.	-0.33 n.s.
U2Net	0.05 n.s.	0.05 n.s.	-0.14 n.s.	-0.24 n.s.	0.05 n.s.	0.14 n.s.	-0.14 n.s.	1.00	0.71 *	0.81 *
TransUNet	-0.05 n.s.	-0.05 n.s.	-0.05 n.s.	-0.14 n.s.	-0.05 n.s.	0.05 n.s.	-0.24 n.s.	0.71 *	1.00	0.90 **
Average	-0.14 n.s.	-0.14 n.s.	-0.14 n.s.	-0.24 n.s.	-0.14 n.s.	-0.05 n.s.	-0.33 n.s.	0.81 *	0.90 **	1.00

FID Clean-FID CLIP-FD RETF.-FD KID CMMD FLD U2Net TransUNet Average

1.0

0.5

0.0

-0.5

-1.0

[Wu et al., 2025c]